

Spin-polarized scanning tunneling microscopy measurement scheme for determining the quantum geometric tensor

Shu-Hui Zhang^{1,*}, Jin Yang,² Ding-Fu Shao³, Jia-Ji Zhu⁴, Wen-Long You⁵, Wen Yang^{6,†} and Kai Chang^{6,7,‡}

¹*School of Physics, Nankai University, Tianjin 300071, China*

²*Beijing Computational Science Research Center, Beijing 100193, China*

³*Key Laboratory of Materials Physics, Institute of Solid State Physics, HFIPS, Chinese Academy of Sciences, Hefei 230031, China*

⁴*School of Science, Chongqing University of Posts and Telecommunications, Chongqing 400061, China*

⁵*College of Physics, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China*

⁶*Center for Quantum Matter, School of Physics, Zhejiang University, Hangzhou 310027, China*

⁷*Institute for Advanced Study in Physics, Zhejiang University, Hangzhou 310027, China*



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The quantum geometric tensor (QGT) embodies the geometry of the eigenstates of a system's Hamiltonian, and its full characterization across diverse quantum systems is essential. However, it is challenging to characterize the QGT of solid-state systems. Here we present an electric scheme to measure the complete QGT of two-dimensional solid-state systems by using spin-polarized scanning tunneling microscopy (STM), in which the spin texture is extracted from geometric amplitudes of Friedel oscillations induced by the intentionally introduced magnetic impurity, and then the QGT is derived from the momentum differential of spin texture. As a canonical spin model, the surface states of a topological insulator offer a promising way to demonstrate the scheme. In a slab of topological insulator, the gapped surface states host complete QGT, i.e., nonvanishing quantum metric and Berry curvature as its symmetric real part and the antisymmetric imaginary part. Thus, a detailed derivation guides the use of the developed scheme to measure the QGT of gapped surface states, even with an external magnetic field. This study opens a new avenue to directly measure the complete QGT of two-dimensional solid-state systems by using spin-polarized STM.

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I. INTRODUCTION

Geometric understanding of various physical systems is remolding modern physics [1]. For a quantum system, there is a Hamiltonian whose solution gives the energy dispersion and eigenstates. The geometric properties of eigenstates are encoded in a quantum geometric tensor (QGT) including the well-established Berry curvature [2–4] and the developing quantum metric [5–7]. The Berry curvature, as the antisymmetric part of the QGT, is well known due to its significant role in topological invariants [8,9], the anomalous Hall effect [10], the spin Hall effect [11], the valley Hall effect [12,13], etc. Conversely, the quantum metric, as the symmetric part of the QGT, is garnering increasing interest due to its key role in orbital magnetic susceptibility [14,15], the exciton Lamb shift in transition-metal dichalcogenides [16], the intrinsic nonlinear Hall effect [17–19], electron-phonon coupling [20], etc. In light of the demand for complete local information in momentum space, there is strong interest in performing the separated measurement of Berry curvature [21–27] or the quantum metric [28,29], and even the full measurement of the QGT [30–36]. Different schemes have been proposed to

measure the QGT of various systems such as photonic lattices [30,33], coupled qubits in diamond [31], and Josephson junctions [32], yet the appropriate scheme for solid state systems is less explored [35–37].

Recently, the complete QGT of exciton-photon modes was directly measured in a two-dimensional high-finesse planar microcavity through the Stokes vectors determined by the polarization intensities of photoluminescence [30]. Specifically, the QGT components can be given through the momentum differential of Stokes vectors:

$$g_{ij} = \frac{1}{4}(\partial_{k_i}\theta\partial_{k_j}\theta + \sin^2\theta\partial_{k_i}\phi\partial_{k_j}\phi), \quad (1a)$$

$$\Omega_z = \frac{1}{2}\sin\theta(\partial_{k_i}\theta\partial_{k_j}\phi - \partial_{k_j}\theta\partial_{k_i}\phi), \quad (1b)$$

for an eigenstate $|u(\mathbf{k})\rangle = [\cos(\theta/2), \sin(\theta/2)e^{i\phi}]^T$ with the polar angle θ and azimuth ϕ . Here, g_{ij} and Ω_z are, respectively, the components of the quantum metric and Berry curvature for the two-dimensional system. In view of the close analogy between the Stokes vector of light and the spin vector of the electron, the QGT of the solid state system is expected to be given by measuring the momentum-space distribution of the spin vector, i.e., spin texture.

The measurement of spin texture bears its own importance [38,39]. For example, for the chiral spin texture of surface states of a topological insulator (TISS), experiments demonstrated its protection from backscattering [38,39] and its evolution with the quantum tunneling coupling in slabs

*Contact author: shuhuizhang@nankai.edu.cn

†Contact author: wenyang@csrc.ac.cn

‡Contact author: kchang@zju.edu.cn

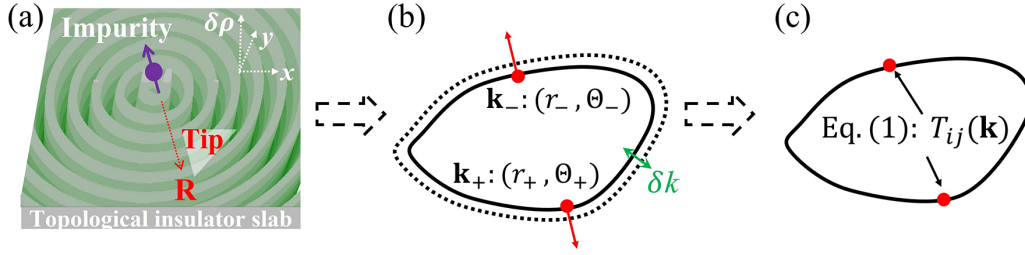


FIG. 1. Schematic STM measurement scheme for the quantum geometric tensor. (a) In a topological insulator slab, spin-polarized Friedel oscillations are induced by the intentionally introduced magnetic impurity, which can be mapped out by the spin-polarized STM measurement. (b) Along any direction $\mathbf{R}$ in real space, the long-range Friedel oscillations are mainly contributed by the coupled backscattering states (red dots) on the constant-energy contour with group velocities (red arrows) parallel to $\pm\mathbf{R}$. For coupled backscattering states at $\mathbf{k}_{\pm}$, the spin vectors characterized by $(r_{\pm}, \Theta_{\pm})$ can be extracted from the geometric amplitudes of Friedel oscillations, which form the spin texture on the constant-energy contour. Based on the energy-resolved STM measurement, the spin textures on different constant-energy (solid and dashed) contours should be provided. (c) If the STM measurement has sufficiently high momentum resolution for spin texture, i.e., sufficiently small δk in (b) depending on the energy resolution of STM measurement, the quantum geometric tensor can be given through the momentum differential of spin texture, i.e., Eq. (1).

[40,41], hexagonal warping degree [42], and magnetic doping [43]. The main technology for measuring the spin texture is limited to spin- and angle-resolved photoemission spectroscopy. Very recently, an experiment [37] used photoemission spectroscopy to directly measure the quantum metric in solids according to Eq. (1). However, the optical measurement of spin texture in the experiment [37] was limited to the $\theta = 0$ case, which cannot consistently provide the complete QGT. Therefore, the general measurement of spin texture is highly anticipated.

Beyond photoemission spectroscopy, scanning tunneling microscopy (STM) is a powerful electric tool for characterizing the electronic structure, with atomic-scale spatial resolution, high energy resolution, and compatibility with the magnetic field [44]. To our knowledge, STM has not yet been able to measure the spin texture. The spin information manifests itself in the geometric amplitudes of spin-polarized Friedel oscillations (FOs) [45] measurable by STM; this echoes the measurement of QGT through the Stokes vector [30]. In this work, we develop an electric scheme to measure the general spin texture ($\theta \neq 0$) and then QGT by using spin-polarized STM. For the sake of universality, the gapped TISS [Fig. 1(a)] of the paradigmatic spin model with a complete QGT is used to demonstrate the scheme. This scheme first extracts the high momentum-resolved spin texture [Fig. 1(b)] from geometric amplitudes of FOs induced by the intentionally introduced magnetic impurity, and then derives the QGT [Fig. 1(c)] from the momentum differential of spin texture according to Eq. (1). This study provides an electric scheme for directly measuring the complete QGT of two-dimensional solid-state systems, and hints at the great potential of the information extraction from the geometric amplitudes of STM.

II. MODEL AND MEASURABLE QGT

To consider a model spin system, i.e., TISS, the general Hamiltonian of the Dirac fermions is given by (see Refs. [46] and Supplemental Material [47])

$$\hat{H} = k_x \sigma_y - k_y \sigma_x + \delta \sigma_z + \mathbf{t} \cdot \mathbf{k}. \quad (2)$$

Here, $\hbar v_F \equiv 1$ is used for dimensionless derivations with v_F for the Fermi velocity, δ denotes the gap opening (e.g., due to the intersurface coupling in a slab [48] or an out-of-plane magnetic field [51]), and $\mathbf{t}$ represents the tilt velocity vector (e.g., due to the in-plane magnetic field [52]). $\sigma_{x,y,z}$ are Pauli matrices for the spin vectors. To solve the Hamiltonian, one can obtain the energy dispersion and wave function, respectively, $E_{\mathbf{k}} = \mathbf{t} \cdot \mathbf{k} + \sqrt{k^2 + \delta^2}$ and $|\psi_{\mathbf{k}}\rangle = |u(\mathbf{k})\rangle e^{i\mathbf{k} \cdot \mathbf{R}}$ (for the conduction band without loss of generality) in which the spinor part is

$$|u(\mathbf{k})\rangle = \frac{1}{\sqrt{1+r^2}} \begin{bmatrix} 1 \\ r e^{i\Theta} \end{bmatrix} \quad (3)$$

with the spinor component $r e^{i\Theta} = (-k_y + i k_x) / (\sqrt{k^2 + \delta^2} + \delta)$. We note here that the expression of Eq. (3) is crucial for generalizing the STM measurement of spin texture from the gapless case [45] to the gapped case [47] and then arriving at the scheme for QGT. To relate two spinor expressions for Stokes vector and TISS to each other, we have $e^{i\theta} = (1 - r^2, 2ir) / (1 + r^2)$ and $\phi = \Theta$, which are helpful for introducing the spin vector as

$$\mathbf{S} \equiv \langle u(\mathbf{k}) | (\sigma_x, \sigma_y, \sigma_z) | u(\mathbf{k}) \rangle \quad (4a)$$

$$= \left(\frac{2r \cos \Theta}{1 + r^2}, \frac{2r \sin \Theta}{1 + r^2}, \frac{1 - r^2}{1 + r^2} \right). \quad (4b)$$

This corroborates the availability of Eq. (1) for the extraction of QGT as shown later.

Usually, by using the Bloch states $|u_n\rangle$ in momentum space, the QGT is defined as [35–37]

$$T_{ij}^n = \langle \partial_{k_i} u_n | \partial_{k_j} u_n \rangle - \langle \partial_{k_i} u_n | u_n \rangle \langle u_n | \partial_{k_j} u_n \rangle. \quad (5)$$

Its symmetric real and antisymmetric imaginary parts correspond to quantum metric $g_{ij}^n = \text{Re}[T_{ij}^n]$ and Berry curvature $\Omega_{ij}^n = -2 \text{Im}[T_{ij}^n]$, respectively. The presence or absence of Berry curvature and quantum metric depends on the symmetry of the Hamiltonian; e.g., $\mathcal{PT}$ symmetry usually enforces vanishing Berry curvature except at some singular points. For the conduction band of the Hamiltonian of Eq. (2), the

QGT includes four independent components [46], which are (omitting of the band index n)

$$\Omega_z = -\frac{\delta}{2(k^2 + \delta^2)^{3/2}}, \quad (6a)$$

$$g_{xx} = \frac{\delta^2 + k_y^2}{4(\delta^2 + k^2)^2}, \quad (6b)$$

$$g_{yy} = \frac{\delta^2 + k_x^2}{4(\delta^2 + k^2)^2}, \quad (6c)$$

$$g_{xy} = -\frac{k_x k_y}{4(\delta^2 + k^2)^2}. \quad (6d)$$

The quantum metric and Berry curvature are both gauge-invariant geometric quantities for the Bloch states under transformation $|u_n\rangle \rightarrow e^{i\varphi}|u_n\rangle$ with $e^{i\varphi}$ an arbitrary phase factor. While Berry curvature is known for its importance in comprehending various anomalous Hall effects [2–4], the quantum metric quantifies the geometric distance between neighboring quantum states and has only recently gained considerable attention [5,7]. According to Eq. (5), the QGT can be formulated if the Bloch states can be identified. Usually, it is a difficult task to identify the Bloch states of a multiband system. But for the two-band system, up to a global undetermined phase factor $e^{i\varphi}$, the Bloch states $|u_n\rangle$ can be characterized by the spin vector [see Eq. (4)] since it is gauge invariant, which then brings about the applicability of Eq. (1) for QGT. In the following, we develop a feasible scheme to measure the spin vector of $|u_n\rangle$ by using spin-polarized STM measurement of FOs induced by the designed single magnetic impurity, and then to characterize the QGT through the momentum differential of spin texture.

III. RESULTS AND DISCUSSIONS

Here, we first introduce the spin-polarized STM scheme to measure the QGT, with a focus on its underlying physics, specific applications and typical features. In light of the paradigmatic role and intensive studies, the gapped TISS is used to demonstrate the proposed scheme, in which both isotropic and tilted cases are considered. In particular, the isotropic case is expected to be verified in the near future, and the tilted case is induced by an in-plane magnetic field [52], which further shows the superiority of STM over the photoemission spectroscopy [37].

A. STM measurement scheme

Using the standard T -matrix approach [45,53], one can obtain FOs characterized by the change of the local density of states [54,55]:

$$\delta\rho_{\alpha\beta}(\varepsilon, \mathbf{R}) = -\frac{1}{\pi} \text{Im Tr}[\mathbf{g}(\varepsilon, \mathbf{R})\mathbf{T}_\alpha\mathbf{g}(\varepsilon, -\mathbf{R})\sigma_\beta], \quad (7)$$

where the T matrix is expressed as

$$\mathbf{T}_\alpha(\varepsilon) = \mathbf{V}_\alpha[1 - \mathbf{g}(\varepsilon, \mathbf{0})\mathbf{V}_\alpha]^{-1}, \quad (8)$$

with $\mathbf{V}_\alpha$ representing the magnetic impurity potential. Here, we use the subscript $\alpha \in \{0, x, y, z\}$ with $\alpha = 0$ and $\alpha \neq 0$ for the spin-unpolarized and spin-polarized impurities or STM tip, respectively, so $\delta\rho_{\alpha\beta}(\varepsilon, \mathbf{r})$ gives β -resolved FOs induced

by the local α -resolved impurity $\mathbf{V}_\alpha = V_0\sigma_\alpha$ of strength V_0 . $\mathbf{g}(\varepsilon, \mathbf{R})$ is the bare Green's function (GF) of the host system for the impurity.

The underlying physics of the proposed scheme can be explained through Fig. 1. In Fig. 1(a), FOs described by Eq. (7) are quantum interference of two propagation waves: one is the propagation $\mathbf{g}(\varepsilon, -\mathbf{R})$ of emitted electron waves from the STM tip σ_β to the impurity, and the other is the propagation $\mathbf{g}(\varepsilon, \mathbf{R})$ of scattered electron waves by the impurity $\mathbf{T}_\alpha$ back to the STM tip. So FOs embody the information of TISS and impurity. If the impurity is designed, e.g., with weak strength and specific polarized direction, one obtains $\mathbf{T}_\alpha \simeq \mathbf{V}_\alpha$. Then, the impurity can be regarded as the other tip, and FOs are the linear responses of TISS measured by two atomic-scale leads [45]. In particular, long-range GFs have the matrix form $\mathbf{g}(\varepsilon, \pm\mathbf{R}) \propto |u(\mathbf{k}_\pm)\rangle\langle u(\mathbf{k}_\pm)|$, in which $\mathbf{k}_\pm$ are the classical momenta of the coupled backscattering states with group velocities parallel to $\pm\mathbf{R}$. The matrix forms of GFs combine the geometric density of states to determine the (geometric) amplitudes of FOs [45]. Thus, it is an inverse problem to solve $|u(\mathbf{k}_\pm)\rangle$ and/or the spin vectors characterized by $(r_\pm, \Theta_\pm)$ from the geometric amplitudes of FOs along any $\mathbf{R}$; see Eq. (3). For any constant-energy contour, the high momentum-resolved spin texture formed by the spin vectors can be given by the atomic-scale resolved STM measurement. Then, the high energy-resolved STM measurement [56,57] favors the high momentum-resolved spin texture in the whole momentum space [see Fig. 1(b)], whose differential gives the QGT [see Fig. 1(c)].

Specific to the system of TISS, impurity-induced FOs can be rewritten into the form:

$$\delta\rho_{\alpha\beta} = \delta\rho_{a,\alpha\beta} \cos[(k_+ + k_-)R] + \delta\rho_{b,\alpha\beta} \sin[(k_+ + k_-)R], \quad (9)$$

whose amplitudes $\delta\rho_{a/b,\alpha\beta}$ can be extracted from the calculations of the exact T -matrix approach or experimental results. If $\delta\rho_{a/b,\alpha\beta}$ satisfy the Born approximation, their spin-resolved components have the analytical expressions ($\alpha \neq 0$ and $\beta \neq 0$) [47]

$$\delta\rho_{a,xx} = C[r_+^2 + r_-^2 + 2r_+r_- \cos(\Theta_- + \Theta_+)], \quad (10a)$$

$$\delta\rho_{a,yy} = C[r_+^2 + r_-^2 - 2r_+r_- \cos(\Theta_- + \Theta_+)], \quad (10b)$$

$$\delta\rho_{a,zz} = C[1 + r_+^2r_-^2 - 2r_+r_- \cos(\Theta_- - \Theta_+)], \quad (10c)$$

$$\delta\rho_{a,xy} = C2r_+r_- \sin(\Theta_+ + \Theta_-), \quad (10d)$$

$$\delta\rho_{b,xy} = C(r_+^2 - r_-^2), \quad (10e)$$

$$\delta\rho_{a,xz} = C[r_-(1 - r_+^2) \cos \Theta_- + r_+(1 - r_-^2) \cos \Theta_+], \quad (10f)$$

$$\delta\rho_{b,xz} = -C[r_-(1 + r_+^2) \sin \Theta_- - r_+(1 + r_-^2) \sin \Theta_+], \quad (10g)$$

$$\delta\rho_{a,yz} = C[r_-(1 - r_+^2) \sin \Theta_- + r_+(1 - r_-^2) \sin \Theta_+], \quad (10h)$$

$$\delta\rho_{b,yz} = C[r_-(1 + r_+^2) \cos \Theta_- - r_+(1 + r_-^2) \cos \Theta_+], \quad (10i)$$

which represent geometric amplitudes inheriting from the geometric density of states [45]. Thus, it is an inverse problem to find C , $r_{\pm}$, and $\Theta_{\pm}$ from $\delta\rho_{a/b,\alpha\beta}$, which are solved analytically for TISS [47]:

$$C = \frac{1}{4}(c_0 - \sqrt{\Lambda + c_0\delta\rho_{a,zz}}), \quad (11a)$$

$$r_{\pm} = \sqrt{\frac{(\delta\rho_{a,xx} + \delta\rho_{a,yy})}{4C} \pm \frac{\delta\rho_{b,xy}}{2C}}, \quad (11b)$$

$$\sin \Theta_{\pm} = \frac{r_{\pm}^2(\delta\rho_{a,yz} \mp \delta\rho_{b,xz}) + (\delta\rho_{a,yz} \pm \delta\rho_{b,xz})}{2Cr_{\pm}(1 - r_{\pm}^2)}, \quad (11c)$$

$$\cos \Theta_{\pm} = \frac{r_{\pm}^2(\delta\rho_{a,xz} \pm \delta\rho_{b,yz}) + (\delta\rho_{a,xz} \mp \delta\rho_{b,yz})}{2Cr_{\pm}(1 - r_{\pm}^2)}, \quad (11d)$$

with

$$c_0 \equiv 2\delta\rho_{a,zz} - 2\sqrt{\Lambda_0 + \Lambda + \delta\rho_{a,zz}^2}, \quad (12a)$$

$$\Lambda_0 \equiv (\delta\rho_{a,xx} + \delta\rho_{a,yy})^2 - 4(\delta\rho_{b,xy})^2, \quad (12b)$$

$$\Lambda \equiv 2(\delta\rho_{a,xz}^2 + \delta\rho_{a,yz}^2) - 2(\delta\rho_{b,xz}^2 + \delta\rho_{b,yz}^2). \quad (12c)$$

As a result, the spin vectors characterized by $(r_{\pm}, \Theta_{\pm})$ are given from the geometric amplitudes of FOs along any $\mathbf{R}$.

For the general applications, the proposed scheme are summarized into three key steps as shown by Fig. 1: (I) inputting $\delta\rho_{\alpha\beta}(\varepsilon, \mathbf{R})$ in real space from experiments or simulations; (II) solving the spin vectors $(r_{\pm}, \Theta_{\pm})$ through the geometric amplitudes of $\delta\rho_{\alpha\beta}(\varepsilon, \mathbf{R})$ for any $\mathbf{R}$ [47], which form the high momentum-resolved spin texture on the constant-energy contour $E_{\mathbf{k}} = \varepsilon$ due to the atomic-scale spatial resolution of STM; (III) deriving QGT through the momentum differential as the difference of spin textures on two constant-energy contours $E_{\mathbf{k}} = \varepsilon$ and $E'_{\mathbf{k}} = \varepsilon + \delta\varepsilon$ with the later given by repeating (I) and (II). STM has the high energy resolution, namely the small $\delta\varepsilon$, that ensures the momentum differential.

The proposed general scheme for the two-band model has two distinct features, i.e., realistic measurement by an electric means and the compatibility with magnetic field. Instead of the developed optical method [36,37], the proposed scheme realizes an electric probe of the QGT of solids systems. Subsequently, we demonstrate this scheme through gapped TISS in the absence and presence of the tilt induced by an in-plane magnetic field. Due to intensive theoretical [55,58–68] and experimental studies [38,69–80], the gapped TISS is the most promising platform to verify our scheme in the near future. In particular, probing the spin texture and/or QGT of fascinating tilted TISS [81–83] is beyond the ability of optical method due to its incompatibility with a magnetic field [44]. We note that the proposed scheme also provides high momentum-resolved measurement of spin texture, which is intrinsically an important issue for STM technology. In addition, we later employ an analytical scheme and a numerical scheme for the amplitude extraction in step (II) through analytical Born approximation and numerical fitting of $\delta\rho_{\alpha\beta}(\varepsilon, \mathbf{R})$ [47], respectively.

B. Isotropic case

Here, we use the proposed three-step scheme for the isotropic case of gapped TISS, in step (I) $\delta\rho_{\alpha\beta}(\varepsilon, \mathbf{R})$ are given through simulations by using the exact T -matrix approach, as

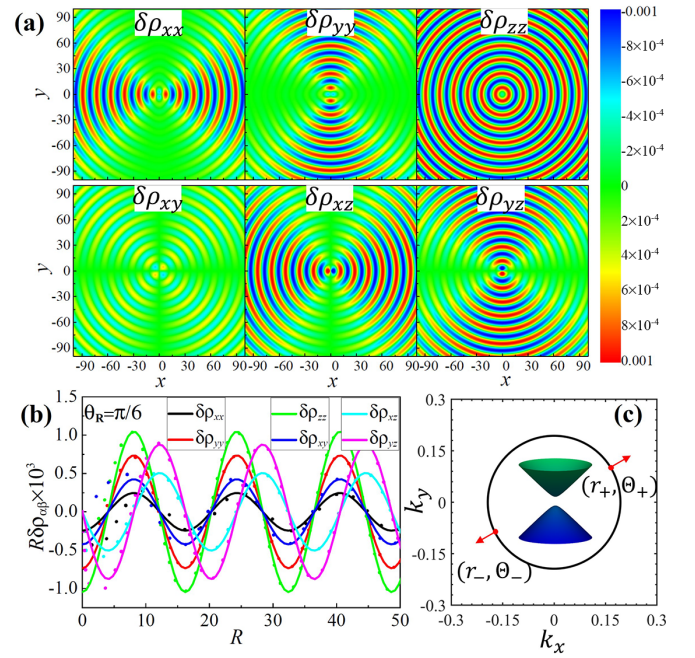


FIG. 2. Spin-polarized Friedel oscillations induced by a single magnetic impurity for isotropic gapped TISS. (a) $\delta\rho_{\alpha\beta}$ calculated by the T -matrix approach. (b) Comparison of $\delta\rho_{\alpha\beta}$ from the T -matrix approach (dotted lines) and the Born approximation (solid lines). (c) For the TISS with isotropic gapped Dirac cone (inset), the constant-energy contour is a circle. The amplitudes of real-space Friedel oscillations along $\mathbf{R}$ in (b) determine the momentum-space spin vectors $(r_{\pm}, \Theta_{\pm})$ of coupled backscattering states (red dots) with group velocities (red arrows) parallel to $\pm\mathbf{R}$. Here, $V_0 = 0.1$, $\varepsilon = 0.2$, and $\delta = 0.05$.

shown by Fig. 2(a). The well-known dimension-determined $1/R$ decay for two-dimensional TISS is used to rescale FOs, i.e., $R\delta\rho_{\alpha\beta}$. Although the Fermi surface of gapped TISS is isotropic [see Fig. 2(c)], the spin-polarized FOs $\delta\rho_{\alpha\beta}$ are typically anisotropic, as expected from the analytical expressions under the Born approximation [47]. In particular, the numerical results from the T -matrix approach and those from the Born approximation for $\delta\rho_{\alpha\beta}$ agree very well even when the oscillations go beyond just one period, as demonstrated by Fig. 2(b).

In step (II), the geometric amplitudes of FOs can be extracted for the coupled backscattering states of Fig. 2(c) with group velocities parallel to $\pm\mathbf{R}$ of Fig. 2(b). The spin vector $(r_{\pm}, \Theta_{\pm})$ should be solved from the geometric amplitudes of FOs, shown by the solid lines in Fig. 3(a) featured by $r_+ = r_-$ and $e^{i\Theta_+} = -e^{i\Theta_-}$ for isotropic gapped TISS [see Eq. (3)]. As a result, the spin vectors in momentum space of coupled backscattering states are given by FOs measurement in real space. Using Fig. 2(a), the high momentum-resolved spin texture on the constant-energy contour is obtained since STM has the atomic-scale spatial resolution.

In step (III), the QGT can be derived from the momentum differential of spin texture. The high momentum-resolved spin textures for the proximate constant-energy contours are realized through the high energy-resolved STM measurement. Typically, the energy resolution of STM can be up to 10 μeV

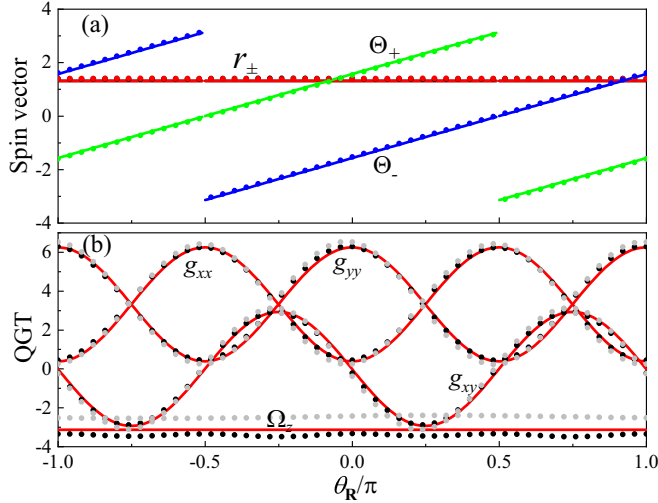


FIG. 3. Spin vector and quantum geometric tensor for gapped TISS. (a) The spin vectors characterized by $(r_{\pm}, \Theta_{\pm})$ for the coupled backscattering states. To consider $V_0 = 0.1$, the results are from the analytical scheme (solid lines) and the numerical scheme (circles). (b) The quantum geometric tensor including four independent components such as Ω_z , g_{xx} , g_{yy} , g_{xy} from the numerical scheme (light grey and black circles) and the analytical scheme (red lines), and $V_0 = 0.1$ is for red and light grey lines while $V_0 = 10^{-3}$ for black ones. Here, $\varepsilon = 0.2$, $\delta = 0.05$, and $\delta k = 10^{-5}$.

at low temperature [56,57]. Adopting $v_F = 10^6$ cm/s and $\delta\varepsilon \propto \delta k$, a proper $\delta k = 10^{-5}$ is used in our calculations. In Fig. 3(b), we note that the analytical scheme is used here for QGT (red lines) and it always reproduces the results from exact expressions (not shown for brevity), i.e., Eqs. (3) and (6). We will discuss the impurity strength effect by using a numerical scheme in the final section.

C. Tilted case

Recently, the tilted TISS induced by an in-plane magnetic field was proposed to explain the planar Hall effect [52]; it attracted wide interest [45,84–87]. However, there is no experimental spectral evidence for the tilted TISS, maybe due to the incompatibility of photoemission spectroscopy with the magnetic field [44]. STM is a promising tool to characterize the tilted TISS, in particular, their QGT, which is our focus. Following the three steps of the scheme, Fig. 4(a) shows spin-polarized FOs for gapped TISS with tilt, and displays the universal $1/R$ decay but becomes more asymmetric compared to Fig. 2(a). Similarly to Fig. 2 (b), $\delta\rho_{\alpha\beta}$ from the T -matrix approach tend towards those from the Born approximation by observing FOs beyond one period in Fig. 4(b). Then, the geometric amplitude can be extracted for the coupled backscattering states of Fig. 4(c) with group velocities parallel to $\pm\mathbf{R}$ of Fig. 4(b). Figure 5 shows the spin texture and QGT by using the analytical scheme (solid lines) and the exact expressions of Eqs. (3) and (6) (dashed lines), which are exactly the same. The numerical scheme is available for the small tilt parameter (not shown) but not for the large tilt parameter, e.g., $\mathbf{t} = (0.3, 0.3)$, because of the intrinsic trouble in calculating the on-site GF of a single Dirac valley [88–92]

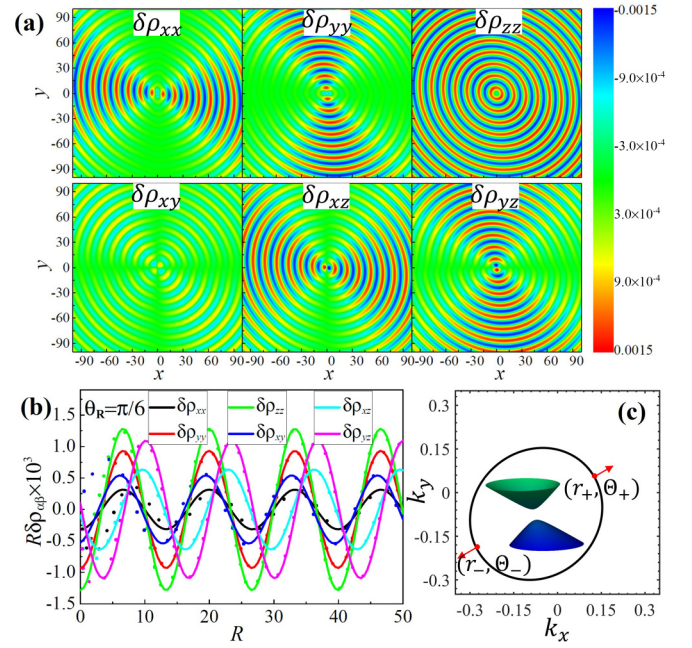


FIG. 4. Spin-polarized Friedel oscillations induced by a single magnetic impurity for tilted gapped TISS. (a) $\delta\rho_{\alpha\beta}$ calculated by the T -matrix approach. (b) Comparison of $\delta\rho_{\alpha\beta}$ from the T -matrix approach (dotted lines) and the Born approximation (solid lines). (c) For the TISS with tilted Dirac cone (inset), the constant-energy contour is an ellipse. The amplitudes of real-space Friedel oscillations along $\mathbf{R}$ in (b) determine the momentum-space spin vectors $(r_{\pm}, \Theta_{\pm})$ of the coupled backscattering states (red dots) with group velocities (red arrows) parallel to $\pm\mathbf{R}$. Here, $\mathbf{t} = (0.3, 0.3)$, $V_0 = 0.1$, $\varepsilon = 0.2$, and $\delta = 0.05$.

enforced by the Nielsen-Ninomiya theorem [93]. This trouble may be tackled for a topological insulator slab by considering two Dirac valleys of the lattice model in future [94].

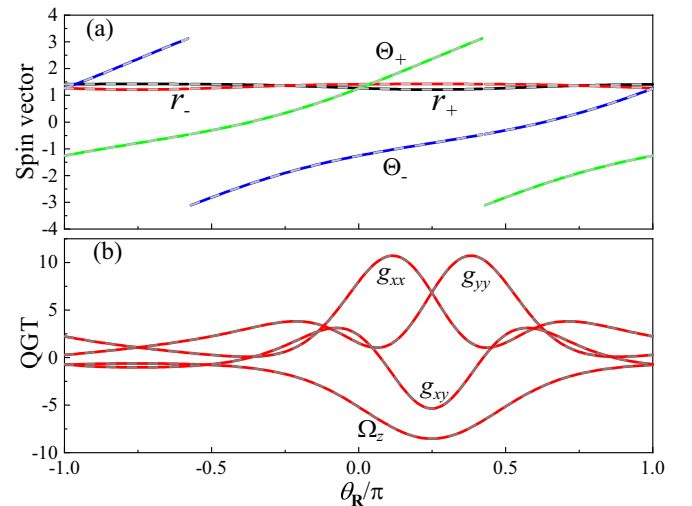


FIG. 5. Spin vector (a) and quantum geometric tensor (b) from the analytical scheme (solid lines) and exact expressions (dashed lines) for tilted gapped TISS. Here, $\mathbf{t} = (0.3, 0.3)$, $V_0 = 0.1$, $\varepsilon = 0.2$, $\delta = 0.05$, and $\delta k = 10^{-5}$.

IV. OUTLOOK AND CONCLUSIONS

The state of the art for spin-polarized STM technology [95–101] favors the extraction of spin texture from FOs induced by a delicately designed magnetic impurity, similar to the experiments for STM measurement of the Berry phase [102,103]. The proposed scheme for the QGT is based on the FOs in real space, which also can give the constant-energy contour [45], and thus bypasses the conventional necessary Fourier transformation [44] and the concomitant signal broadening due to finite image area. To realize the proposed scheme, the magnetic impurity on TISS with magnetization directions $\sigma_{x,y,z}$ and weak V_0 is necessary. For the former, use of a magnetic field is promising for fixing the magnetization directions of the magnetic impurity, and its effect on the electronic Hamiltonian of Eq. (2) can be incorporated through renormalizing the parameters for gap opening and tilt [46]. For the latter, by using the numerical scheme, $V_0 = 0.1$ favors the effective extraction of the spin texture [circles in Fig. 3(a)] and QGT [light gray lines in Fig. 3(b)] while $V_0 = 0.001$ works better [black lines for QGT in Fig. 3(b)] since weak V_0 implies the accuracy of the Born approximation. If one regards the magnetic impurity as a tip, dual-probe STM should provide an alternative way to characterize the spin texture [104] and then the QGT. Furthermore, the lifetime of TISS can be up to $\sim$ ps [105]; this ensures insignificant scattering by other defects, particularly the nonmagnetic impurities due to backscattering suppression of TISS [38]. Even considering the finite lifetime of TISS, it fringes FOs with an exponential decay [106] in addition to the dimension-determined decay $1/R$, which can be well incorporated into our scheme through the amplitude extraction.

The key of the developed scheme is the information extraction from the geometric amplitudes of the STM measurement; this can be extended to the other measurements, e.g., photoemission spectroscopy [107]. This information extraction is actually to solve an inverse problem, which generally requires a numerical treatment, e.g., for the *ab initio* electronic structure with an effective two-band description near the Fermi level [37], although it is analytical for our considered model

Hamiltonian for gapped TISS. And the related dataset is rather huge, e.g., energy-resolved and space-resolved FOs in our scheme. As a result, data processing technology based on machine learning [108] is expected to be developed. In addition, it is interesting to develop the scheme to locally measure the pseudospin texture [109] and orbital texture [110], and the relevant QGT.

To conclude, it is a challenging problem to measure the QGT of solid state systems. To this aim, we develop a spin-polarized STM measurement scheme for the QGT of two-dimensional solid-state systems, which includes the extraction of spin texture from geometric amplitudes of FOs induced by the intentionally introduced magnetic impurity and then the momentum differential of spin texture for the QGT. Using the TISS as an example, we showcase the application of the developed scheme. This study theoretically realizes the electric measurement of the QGT of two-dimensional solid-state systems and highlights the great potential of STM to obtain the geometric information of electronic structure.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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